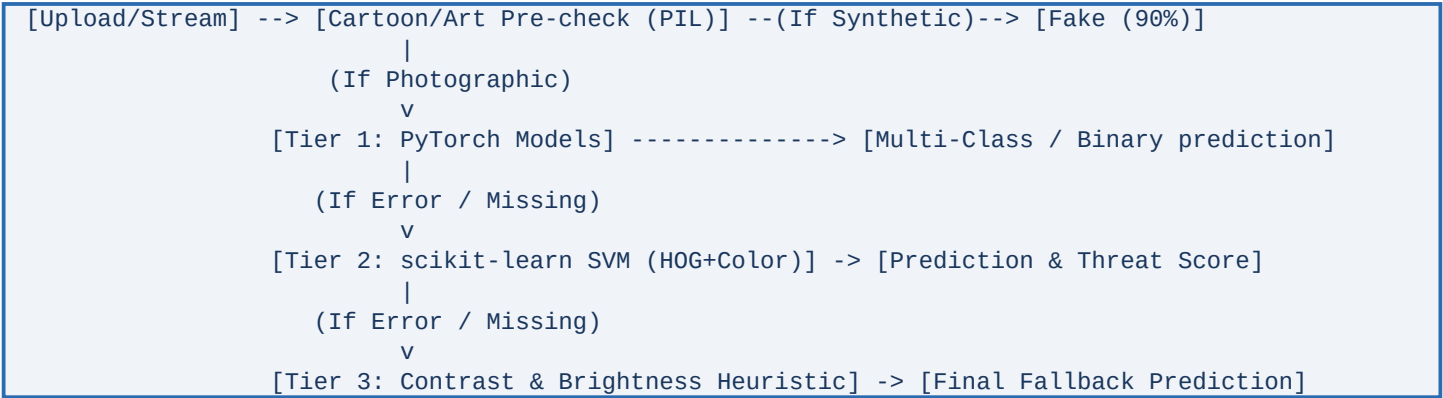


1. Detection Pipeline Overview

Verifixia operates a multi-tier detection pipeline designed to maximize processing efficiency and accuracy. Each incoming request passes through automated pre-checks before hitting the core deep learning models. In the event of an infrastructure or PyTorch execution failure, the system seamlessly cascades down through alternative classifiers to ensure continuous uptime.



2. Multi-Class Image Detector (Primary PyTorch Model)

The primary deep learning model is a custom 3-class classifier designed to separate authentic media from different types of manipulations (AI synthesis vs. classical splicing/swaps).

Architecture	Custom ResNet backbone integrated with Squeeze-and-Excitation (SE) Attention
Target Classes	3 classes: 0 = Real, 1 = Deepfake, 2 = AI-Generated
Total Parameters	12,975,675
Trainable Parameters	12,975,675
Input Dimensions	3 channels x 299 x 299 pixels
Classification Head	Linear bottleneck head (1024 -> 512 -> 256 -> 3 classes) with Dropout
Peak Validation Accuracy	89.09% (trained over 80 epochs)
Training Weight File	models/multiclass_detector.pth

3. Binary Image Deepfake Detector (Xception Fallback)

For binary verification, the platform contains an alternative Xception-based deep learning model trained strictly on binary classification. It utilizes a similar custom attention architecture.

Architecture	Xception-based CNN / DeepfakeDetector with SE Attention blocks
Target Output	Binary prediction: Real (0) or Fake/Deepfake (1)
Total Parameters	12,975,161
Input Dimensions	3 channels x 299 x 299 pixels
Classification Head	Sequential Dense layers (1024 -> 512 -> 256 -> 1) with Sigmoid activation
Training Weight File	models/xception_deepfake.pth

4. Video Deepfake Detector (DeeperForensics Pipeline)

The video detection pipeline handles dynamic content using a frame-sample aggregation strategy, integrating facial localization and temporal analysis.

Face Extractor	OpenCV Haar Cascades (frontal face detection)
Frame Sampling	Extracts up to 25 evenly-spaced frames from the video sequence
Decision Threshold	0.5 (loaded from models/deeperforensics_info.json)
Evaluation Dataset	DeeperForensics-1.0 local evaluation subset
Training Epochs	30 epochs
Best Validation Accuracy	100.0% (on local validation split)
Aggregation Function	Frame-wise average of binary confidence scores

5. Fallback Classifiers (Tiers 2 & 3)

To guarantee high system availability and zero downtime on low-resource environments (such as basic CPU nodes), Verifixia integrates non-deep-learning classifiers.

Tier 2 Fallback: scikit-learn SVM

A classical Support Vector Machine (SVM) model trained on hand-crafted visual features. It runs without a GPU.

Classifier Type	Support Vector Machine (SVM) with RBF (Radial Basis Function) Kernel
Feature Set	Histogram of Oriented Gradients (HOG) + RGB/YCbCr color histograms + FFT spectrum
Dimensionality	Concatenated multi-space feature vector normalized via StandardScaler
Input Size	128 x 128 pixels (downscaled for fast feature extraction)
Saved Package File	models/deepfake_sklearn.pkl

Tier 3 Fallback: Heuristic Analysis

If both PyTorch and scikit-learn modules fail, the system runs raw pixel statistics using standard Pillow libraries. It evaluates absolute brightness offsets and standard deviation contrast measurements, blending them with a slight random fluctuation to construct a provisional confidence indicator.

Cartoon & Synthetic Art Detector (Pre-processing Filter)

Before feeding an image to PyTorch/SVM, a dedicated heuristic filter checks color variety and channel uniformity. Since realistic deepfake models perform poorly on illustrated media, drawings are immediately classified as Synthetic/Fake with 90% confidence, saving CPU/GPU resources.